

EP 03 The Focal Distance of Lenses

Objective

To measure and calculate the focal distance of thin lenses including convex and concave and to clarify the different methods to measure the focal length.

Principle

Converging lens

Plane Mirror Method

The light path is reversible. Thus we can arrange a plane mirror behind the lens and adjust the illuminated objects until the refracted rays are normal to the plane mirror.

Lens Formula Method

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

where u is the distance from the object to the lens, v is the distance from the image to the lens, and f is the focal length of the lens. So we can measure several value of u, v to calculate the value of f .

Two-Position Method

By changing the position of the lens until it processes another sharp image.

$$f = \frac{D^2 - l^2}{4D}$$

where D is the distance from the object to the screen, l is the distance between two positions of the lens.

Note that the real image can not be obtained if the object is inside the principal focus or the distance between the object and the screen is less than $4f$. Thus in the method the separation of the object and screen must exceed $4f$.

Diverging Lens

Auxiliary Converging Lens Method

Since the diverging lens normally forms a virtual image, which cannot be located by a screen and is not easily found by pin or no parallax method, we need the help of a converging lens.

Initially the diverging lens is absent and the image of the converging lens is located and notes its position, called P_1 . Then place the diverging lens between the converging lens and P_1 , then the diverging lens forms a real image of P_1 at P_2 , thus u is the distance between the diverging lens and P_1 and v is the distance between the diverging lens and P_2 . Then we can calculate the f by upper formula.

Procedures

Data

Converging Lens

Plane mirror method

$f =$

Lens Formula Method

- table 1

table1

	$x1$	$x2$	$x3$	u	v	f
1						
2						
3						
4						
5						
平均						

- Plot the graph $1/v$ against $1/u$.

Two-position method

	$x1$	$x2$	$x3$	$x4$	D	l	f
1							
2							
3							
4							
5							
平均							

Diverging lens

Auxiliary converging lens method

	$x1$	$x2$	$x3$	u	v	f
1						
2						
3						
平均						

Pre-Questions

The other methods to measure the focal length of converging or diverging lens.

Questions

- What should be prepared before you measuring the focal length
- which method is more accurate and why.